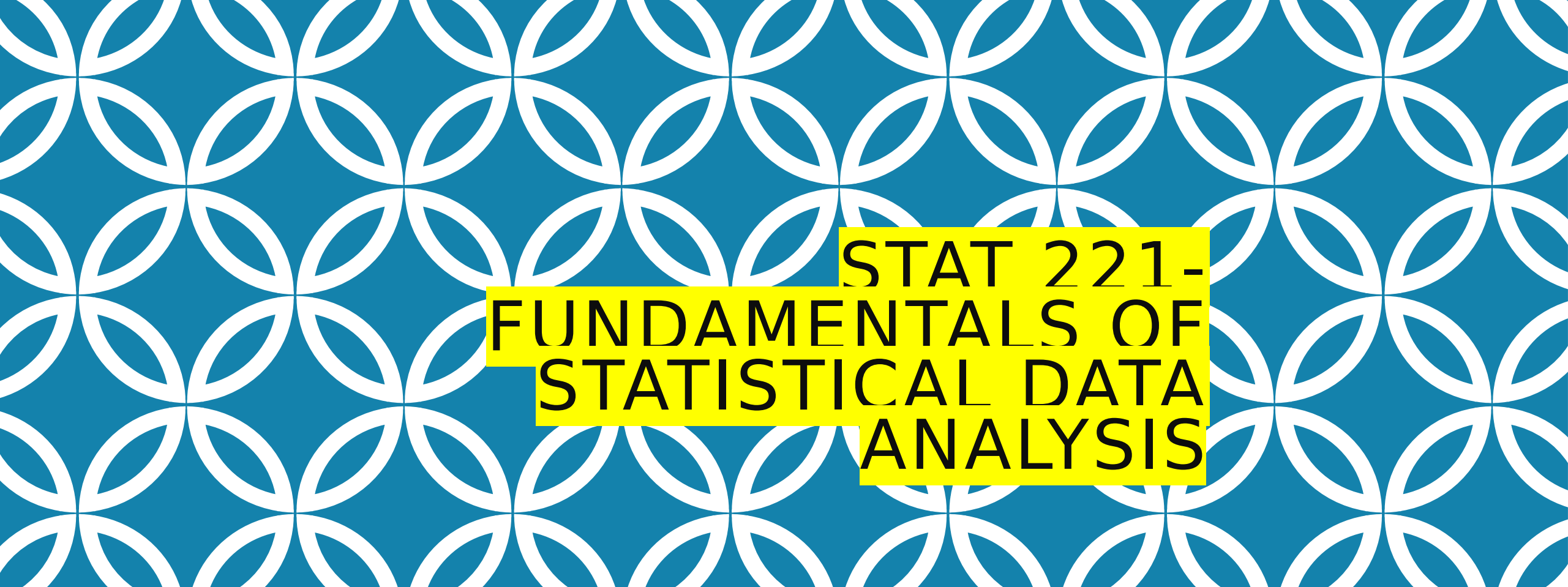
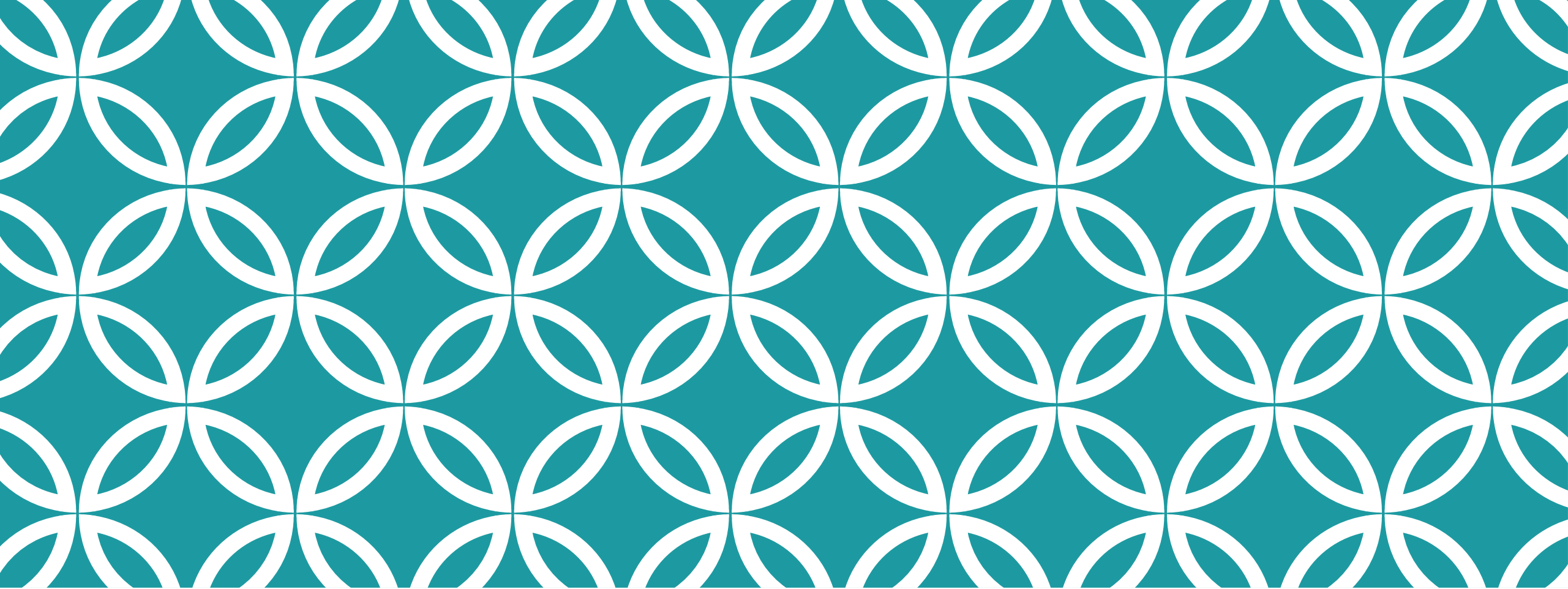




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-14: Correlation and Regression

Shivakumar Jolad



CORRELATION AND REGRESSION |

DATA

Quantitative explanatory variable X

Quantitative response variable Y

Objective: To quantify the *linear* relationship between X and Y

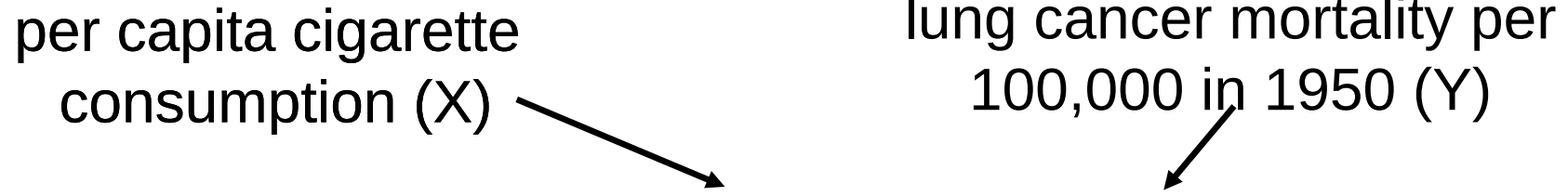
Table 14.1 Synonyms for *explanatory variable* and *response variable*.

Explanatory Variable	→	Response Variable
X	→	Y
independent variable	→	dependent variable
factor	→	outcome
treatment	→	response
exposure	→	disease

ILLUSTRATIVE DATA (DOLL, 1955)

per capita cigarette
consumption (X)

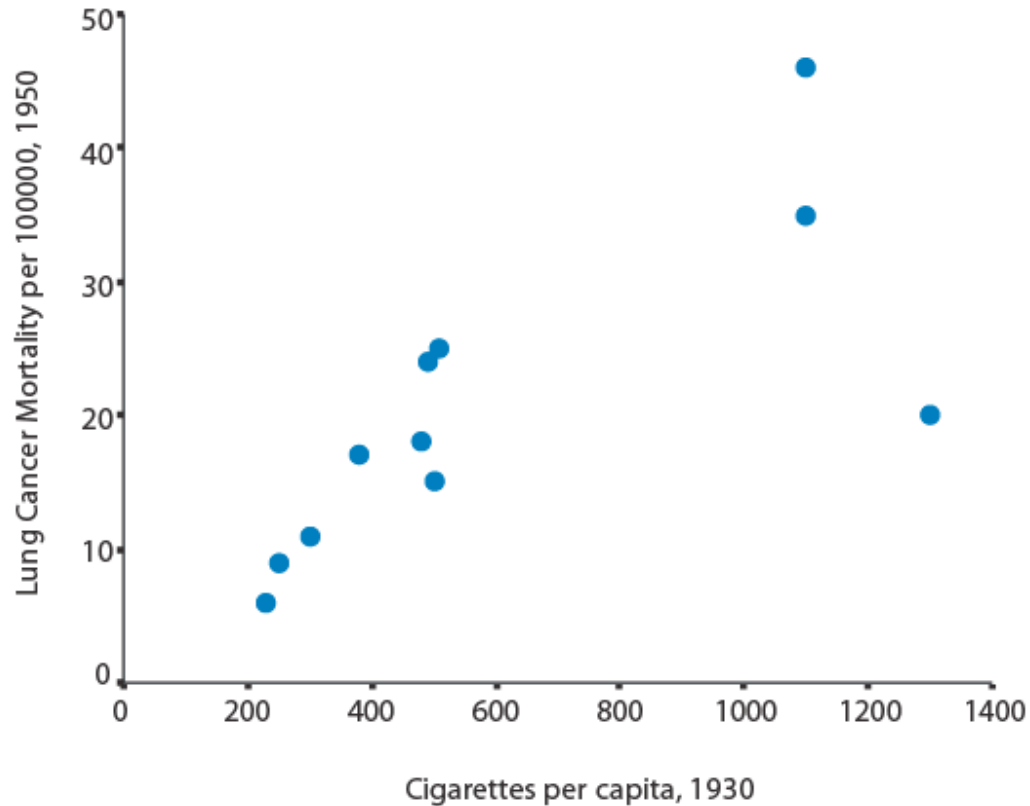
lung cancer mortality per
100,000 in 1950 (Y)



COUNTRY	CIG1930	LUNGCA
USA	1300	20
Great Britain	1100	46
Finland	1100	35
Switzerland	510	25
Canada	500	15
Holland	490	24
Australia	480	18
Denmark	380	17
Sweden	300	11
Norway	250	9
Iceland	230	6

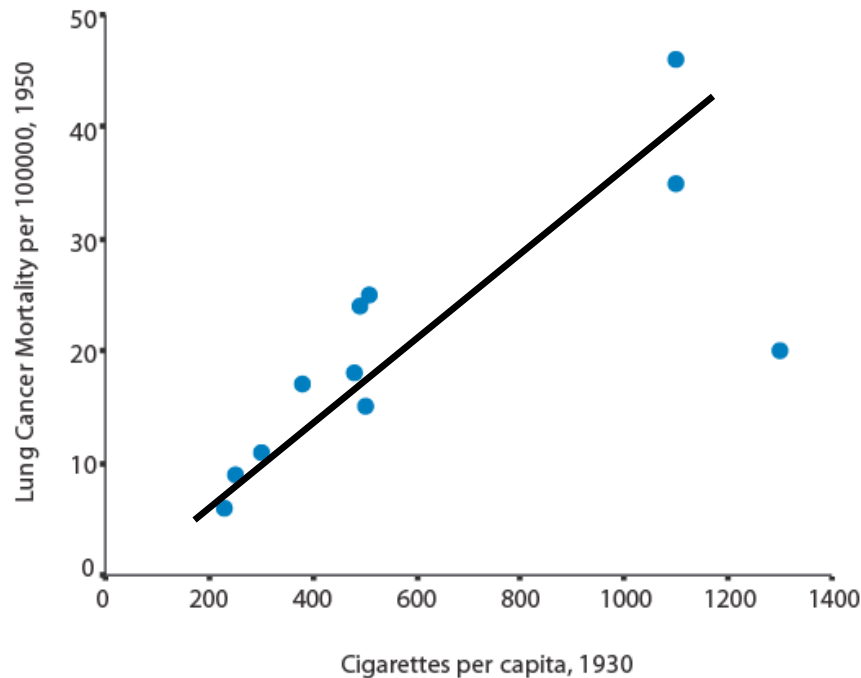
$n = 11$

SCATTERPLOT



Assess:
Form
Direction of
association
Outliers
Strength of
relation

DOLL, 1955



Form: linear

Direction: positive association

Outlier?

Strength: difficult to determine by eye

CORRELATION COEFFICIENT r

r = Pearson's product-moment correlation coefficient

Measures degree to which X and Y “go together”

Always between -1 and 1

$r \approx 0 \Rightarrow$ no correlation

$r > 0 \Rightarrow$ positive correlation

$r < 0 \Rightarrow$ negative correlation

Closer r is to 1 or -1 , the **stronger** the correlation



Karl Pearson
1857 - 1936

CORRELATIONAL DIRECTION AND STRENGTH



Perfect positive correlation $r = 1.0$



Strong positive correlation $r = 0.9$



Moderate positive correlation $r = 0.5$



Perfect negative correlation $r = -1.0$



Strong negative correlation $r = -0.9$



Moderate negative correlation $r = -0.5$

INTERPRETATION OF R

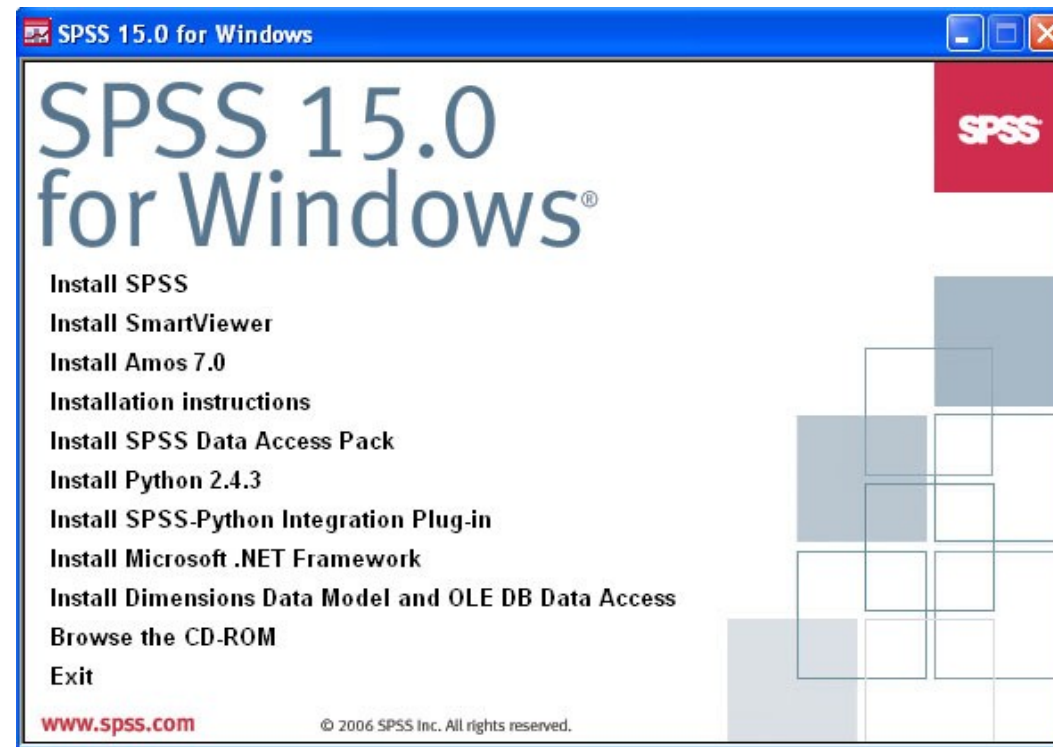
Direction of association: positive, negative, ~ 0

Strength of association

- close to 1 or -1 $\Rightarrow$ “strong”
- close to 0 $\Rightarrow$ “weak”
- guidelines
 - if $|r| \geq .7 \Rightarrow$ say “strong”
 - if $|r| \leq .3 \Rightarrow$ say “weak”

CALCULATING *R*

By hand, calculator or computer program
We opt for latter



SPSS OUTPUT

SPSS > Analyze > Correlate > Bivariate

Correlations

		Cig Consumption per capita, 1930	Lung Cancer Mortality per 100000, 1950
Cig Consumption per capita, 1930	Pearson Correlation	1	.737**
	Sig. (2-tailed)		.010
	N	11	11
Lung Cancer Mortality per 100000, 1950	Pearson Correlation	.737**	1
	Sig. (2-tailed)	.010	
	N	11	11

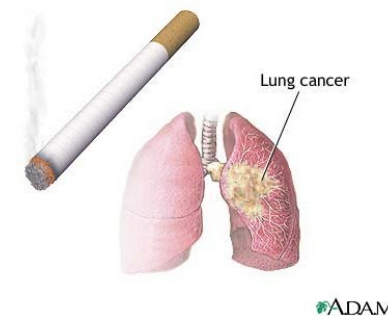
** . Correlation is significant at the 0.01 level (2-tailed).

$r = 0.74$ indicates a strong, positive association

COEFFICIENT OF DETERMINATION (R^2)

Square the correlation coefficient $\Rightarrow r^2 =$ proportion of variance in Y mathematically explained by X

Illustrative data: $r^2 = 0.737^2 = 0.54 \Rightarrow 54\%$ of variance in lung cancer mortality is mathematically explained per capita smoking rates



CAUTIONS

Outliers

Non-linear relations

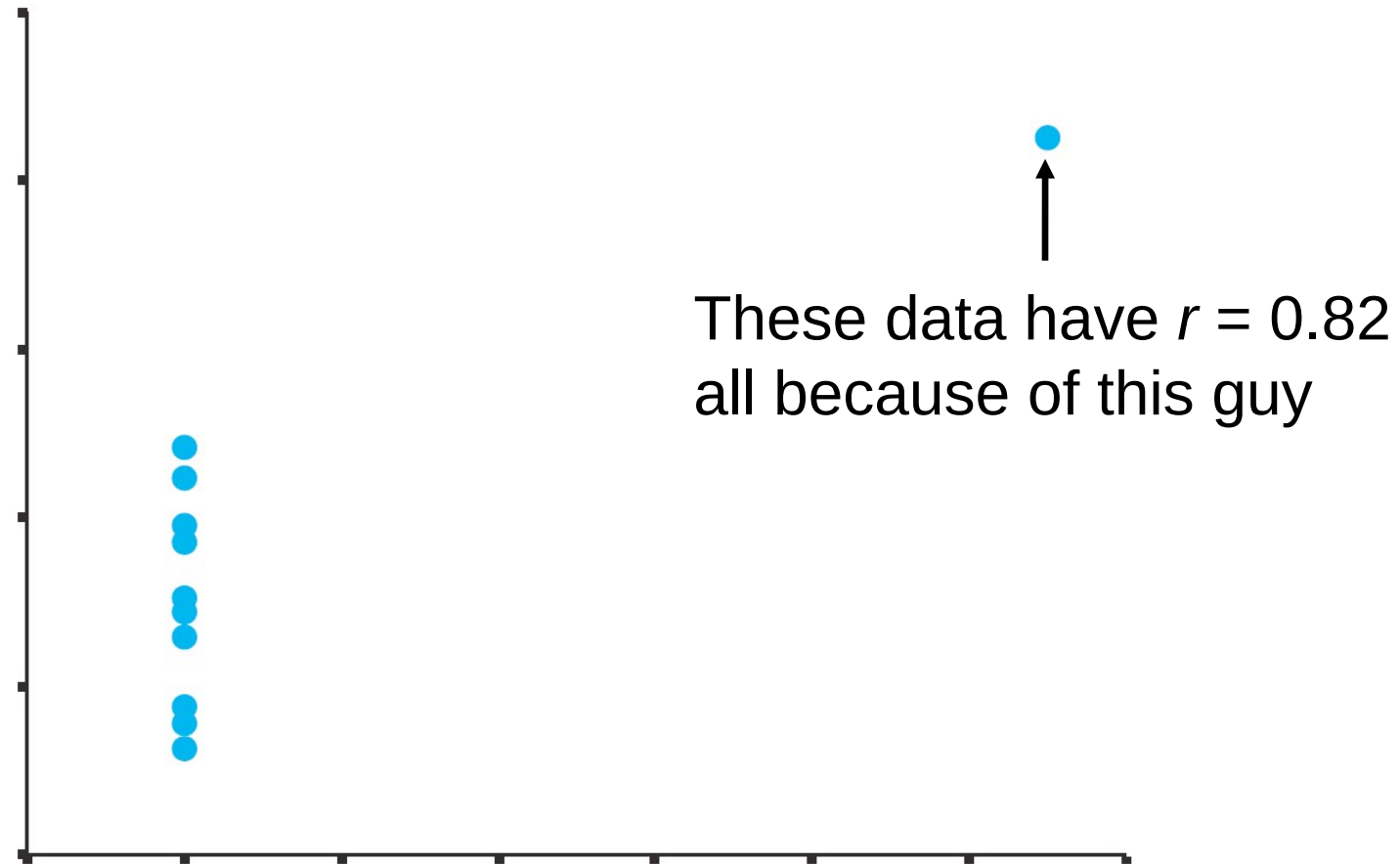
Confounding (correlation is NOT causation)

Randomness

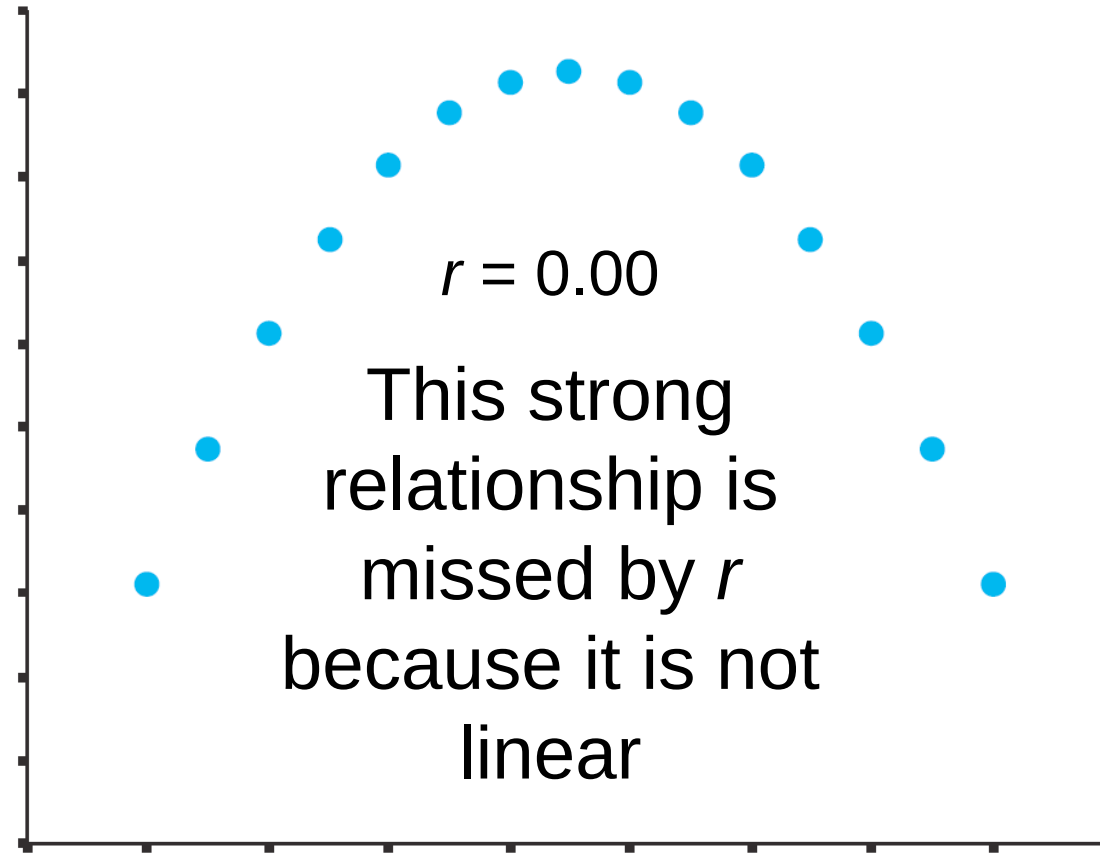


OUTLIERS

Outliers can have profound influence on r

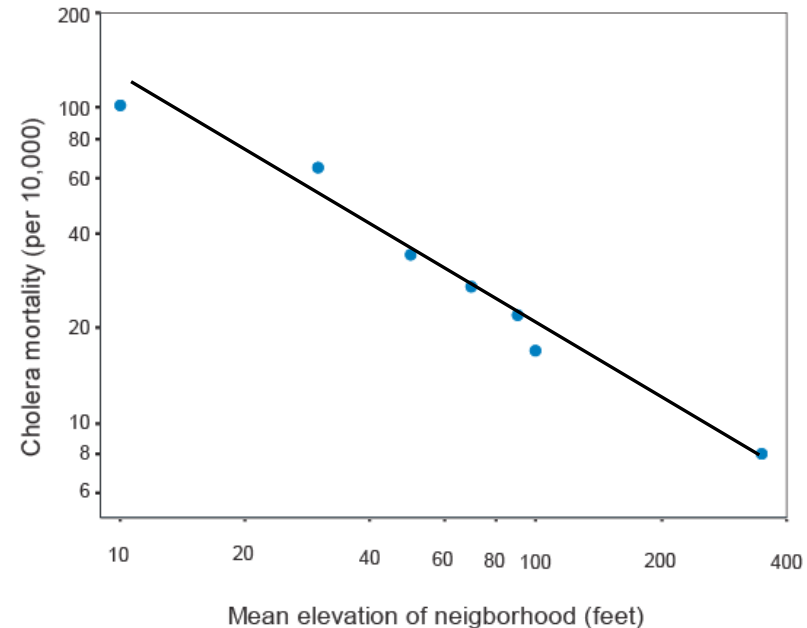


LINEAR RELATIONS ONLY



CONFOUNDING CORRELATION $\neq$ CAUSATION

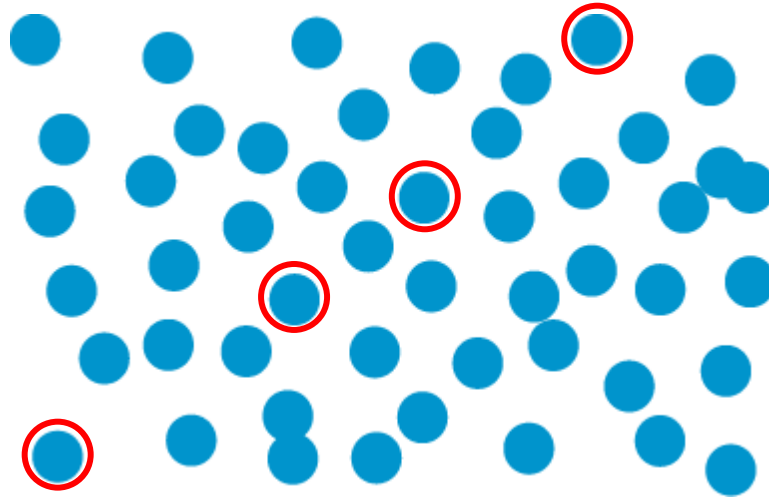
William Farr showed this strong negative correlation between cholera mortality and elevation above sea level in defense of miasma theory



However, he failed to account for the fact that people who lived at low elevations were more likely to drink from contaminated water sources ($\therefore$ confounding)

DON'T BE FOOLED BY RANDOMNESS

Selection of specific data points would result
in a false correlation



HYPOTHESIS TEST

Test the claim $H_0: \rho = 0$

where $\rho \equiv$ correlation coefficient parameter

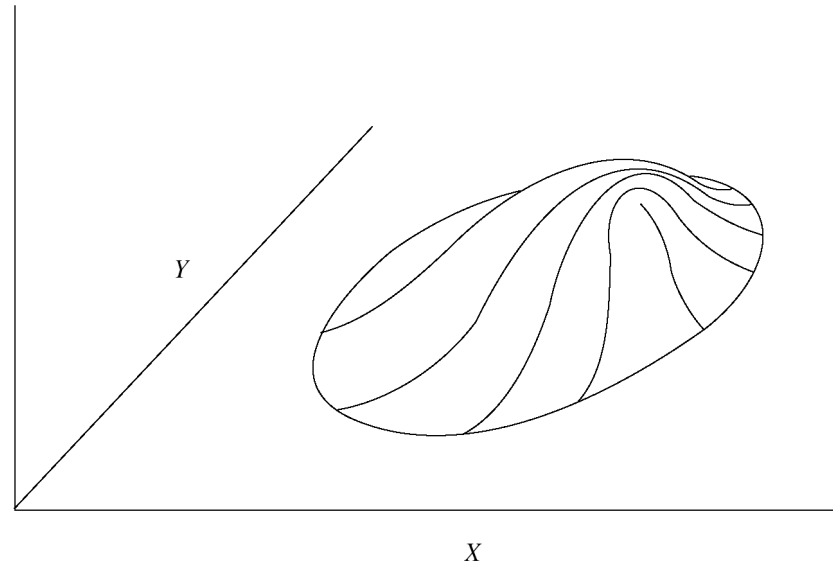
SPSS > Analyze > Correlate > Bivariate output:

Correlations			
		Cig Consumption per capita, 1930	Lung Cancer Mortality per 100000, 1950
Cig Consumption per capita, 1930	Pearson Correlation	1	.737**
	Sig. (2-tailed)		.010
	N	11	11
Lung Cancer Mortality per 100000, 1950	Pearson Correlation	.737**	1
	Sig. (2-tailed)	.010	
	N	11	11

** . Correlation is significant at the 0.01 level (2-tailed).

$\therefore P = .010$ (two-sided) $\Rightarrow$ reliable evidence against H_0
 $\Rightarrow$ the correlation is statistically significant

BIVARIATE NORMALITY



Strictly speaking: P -value requires Normality of the joint distribution of X and Y (“bivariate Normality”)

§14.4. REGRESSION

Regression model (equation for line):

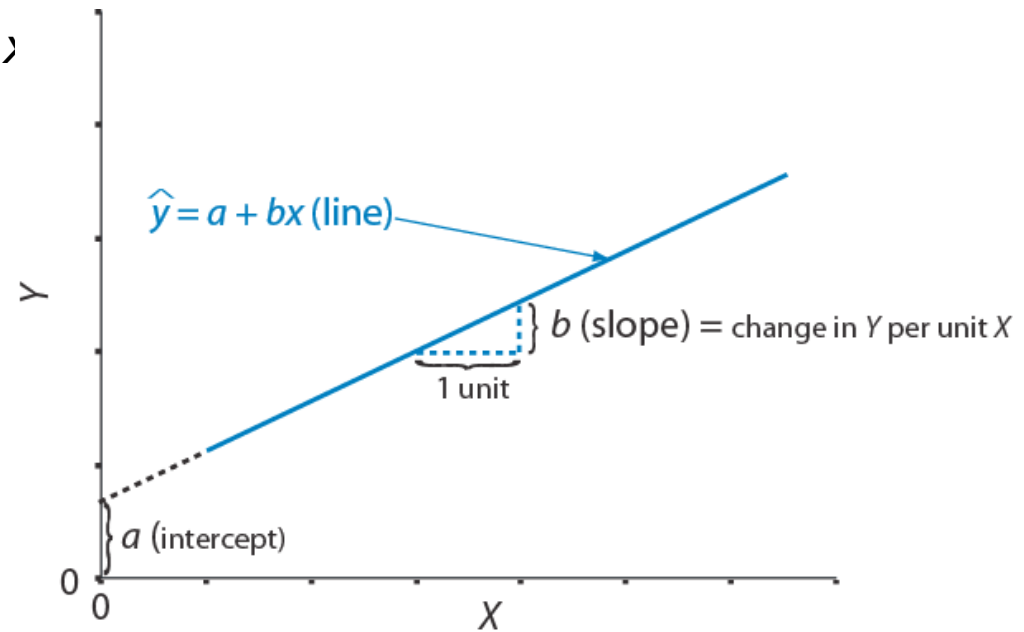
$$\hat{y}_i = a + b \cdot X_i$$

where

$\hat{y}_i \equiv$ predicted value of Y at x_i

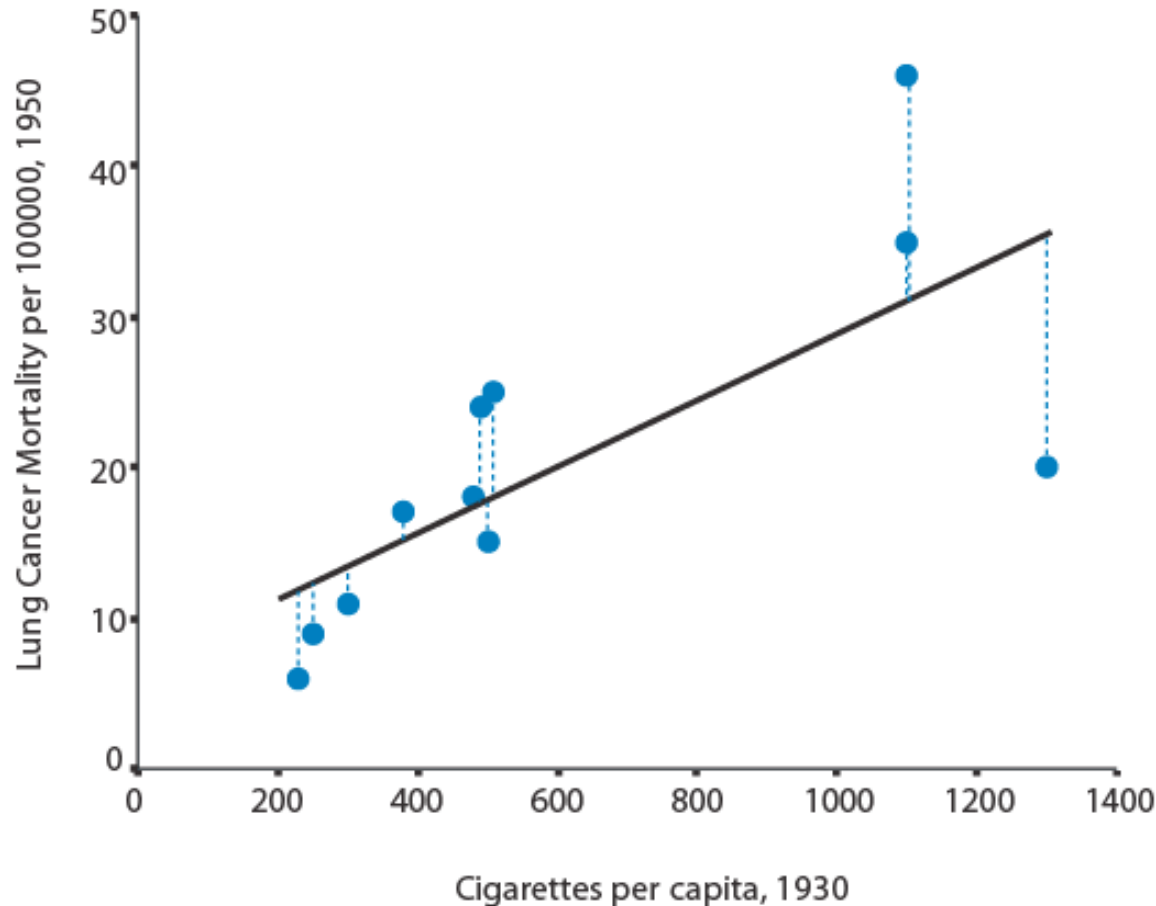
$a \equiv$ intercept coefficient

$b =$ slope coefficient



LEAST SQUARES LINE

Residual $\equiv$ distance of data point from regression line (dotted)



The best fitting line minimizes the residuals

Determine a and b of best fitting line via formula, calculator, or computer.

COEFFICIENT BY SPSS

ANALYZE > REGRESSION > LINEAR

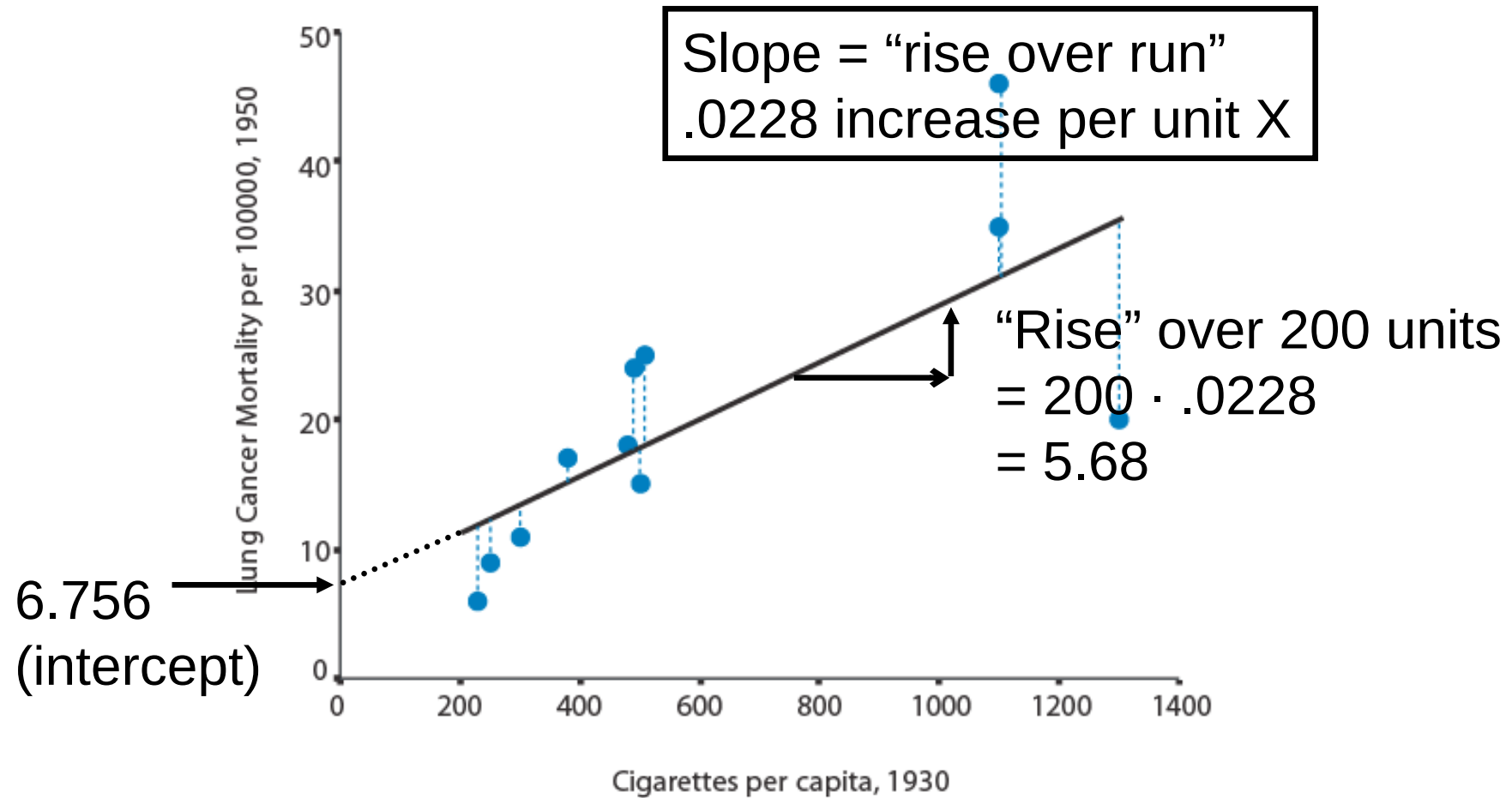
		Intercept estimate (a)		Slope estimate (b)			
		Coefficients ^a					
		Unstandardized Coefficients		Standardized Coefficients			
Model		B	Std. Error	Beta		t	Sig.
1	(Constant)	6.756	4.906			1.377	.202
	CIG1930	2.284E-02	.007	.737		3.275	.010

a. Dependent Variable: MORTALIT

Regression line:

$$\hat{y} = 6.756 + 0.02284 \cdot X$$

$$\hat{Y} = 6.756 + 0.02284 \cdot X$$



POPULATION REGRESSION MODEL

$$y_i = \alpha + \beta x_i + \epsilon_i$$

where

$\alpha \equiv$ intercept parameter

$\beta \equiv$ slope parameter

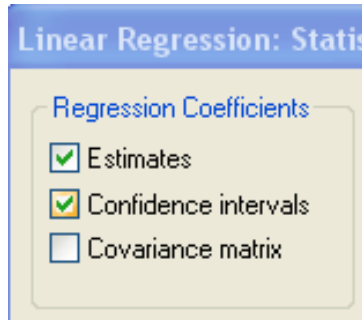
$\epsilon_i \equiv$ residual error, observation i

Objective:

To estimate β with $(1 - \alpha)100\%$ confidence

CI FOR B

ANALYZE > REGRESSION > LINEAR > STATISTICS



SPSS statistics options
Dialogue box

Model		95% Confidence Interval for B	
		Lower Bound	Upper Bound
1	(Constant)	-4.342	17.854
	cig1930	.007	.039

95% CI for β

95% CI for β (.007 to .039)

TESTING $H_0: B = 0$

P value

t_{stat}

		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
Model		B	Std. Error	Beta		
1	(Constant)	6.756	4.906		1.377	.202
	cig1930	.023	.007	.737	3.275	.010

$$df = n - 2 = 11 - 2 = 9$$

$\therefore P = .010 \Rightarrow$ evidence against H_0 is good
 $\Rightarrow$ the slope is statistically significant

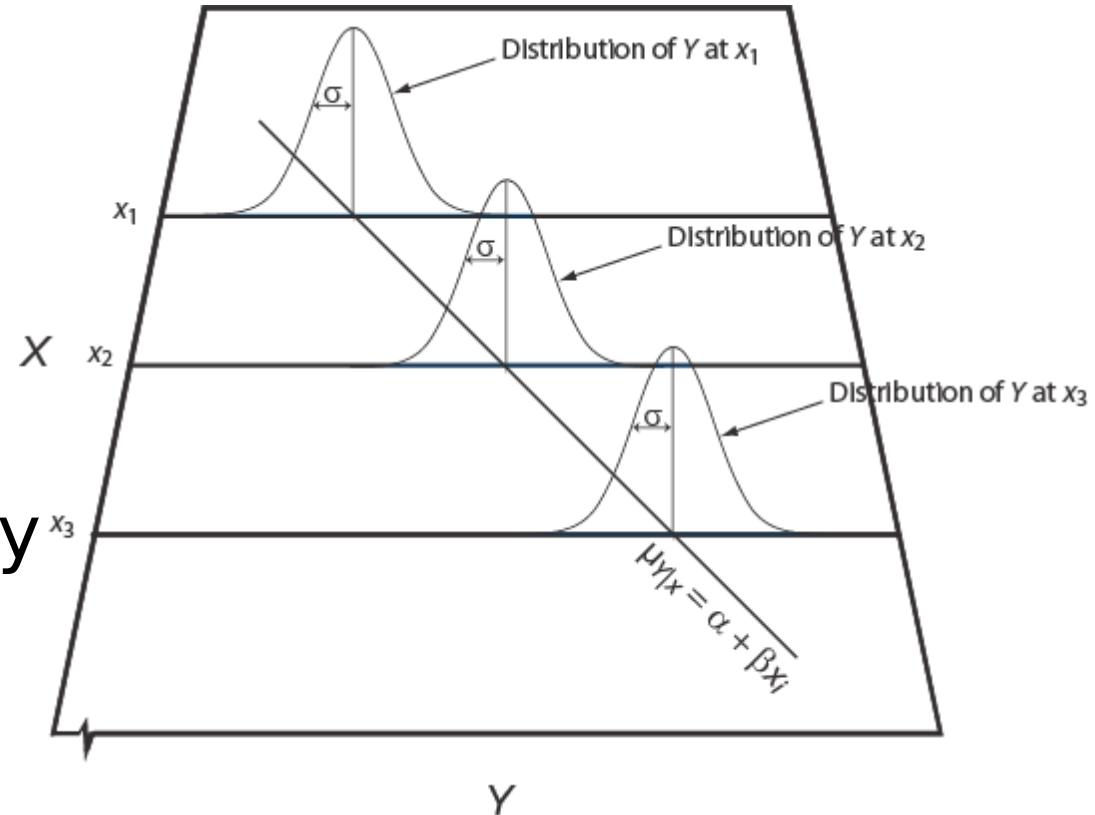
CONDITIONS FOR REGRESSION INFERENCE

Linearity

Independent
observations

Normality

Equal variance
(homoscedasticity
)

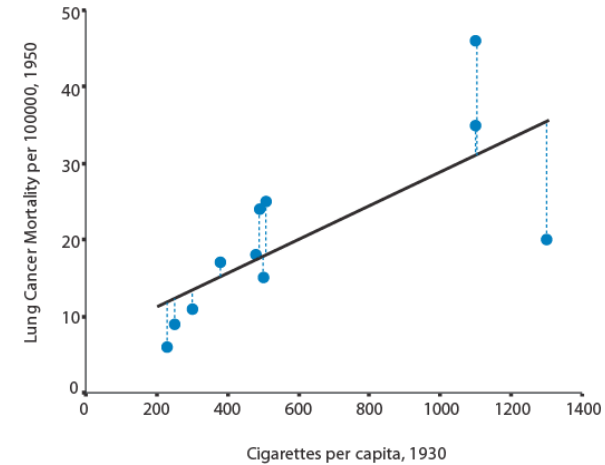


ASSESSING L.I.N.E

Inspect scatterplot for linearity

Inspect residuals for

- linearity
- Normality
- equal variance



i	COUNTRY	X	Y	predicted $\hat{y}$	residual $y_i - \hat{y}_i$
1	USA	1300	20	36.453	-16.453
2	GrBritain	1100	46	31.884	14.116
3	Finland	1100	35	31.884	3.116
4	Switzerland	510	25	18.406	6.594
5	Canada	500	15	18.178	-3.178
6	Holland	490	24	17.950	6.050
7	Australia	480	18	17.721	0.279
8	Denmark	380	17	15.437	1.563
9	Sweden	300	11	13.609	-2.609
10	Norway	250	9	12.467	-3.467
11	Iceland	230	6	12.010	-6.010

ASSESSING CONDITIONS

residual
 $y_i - \hat{y}_i$

-16.453
14.116
3.116
6.594
-3.178
6.050
0.279
1.563
-2.609
-3.467
-6.010

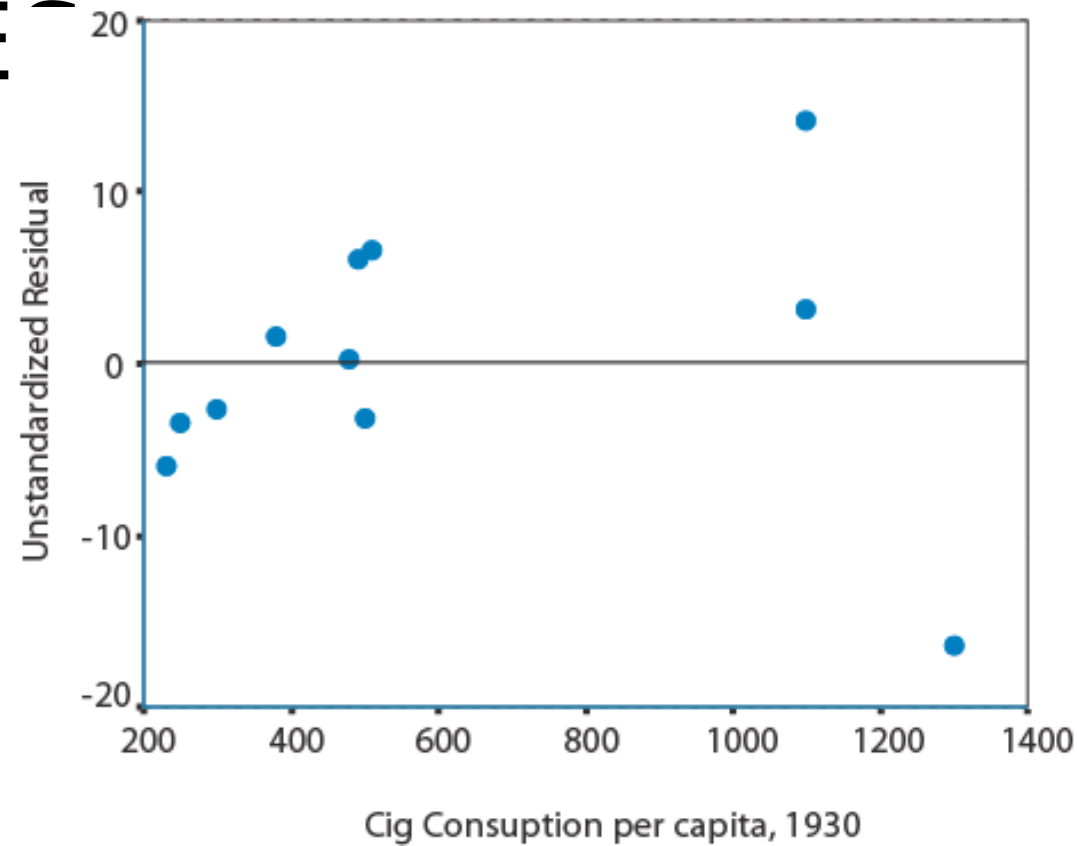
⇒

-1 | 6
-0 | 2336
0 | 01366
1 | 4
x10

⇒

no major
departures
from
Normality

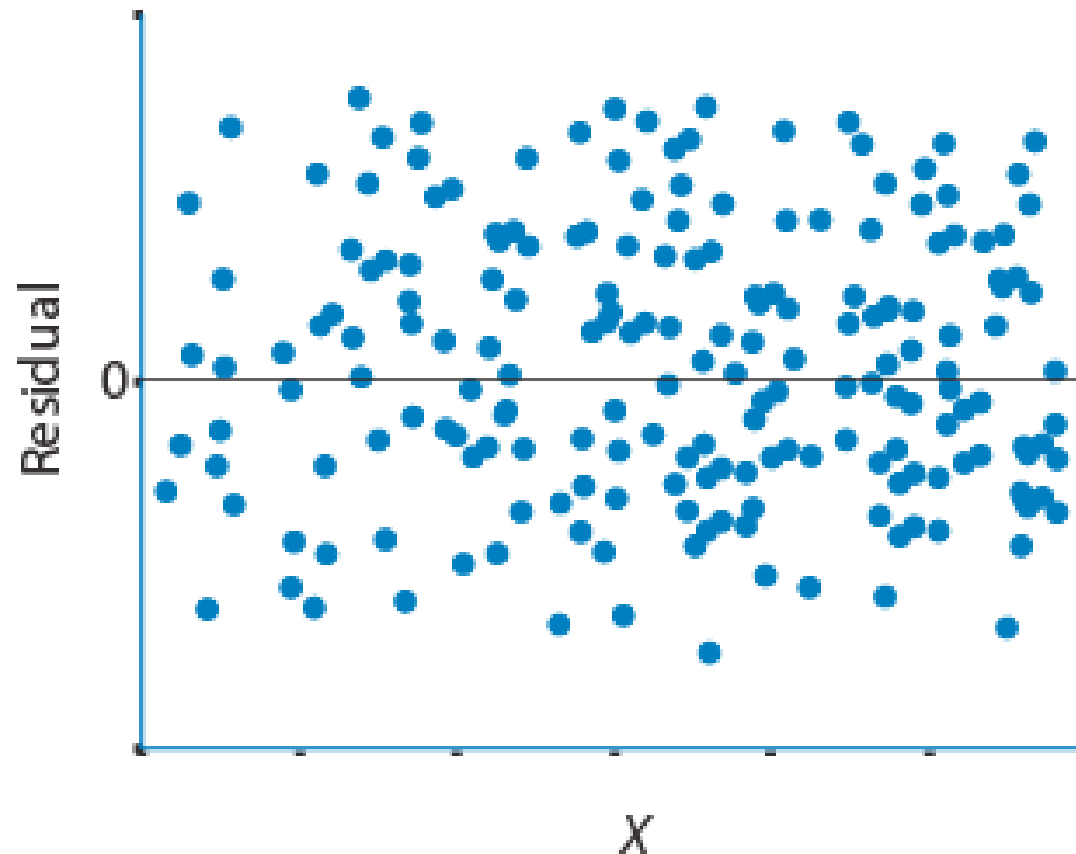
RESIDUAL PLOTTED AGAINST X VALUE



Data too sparse to assess

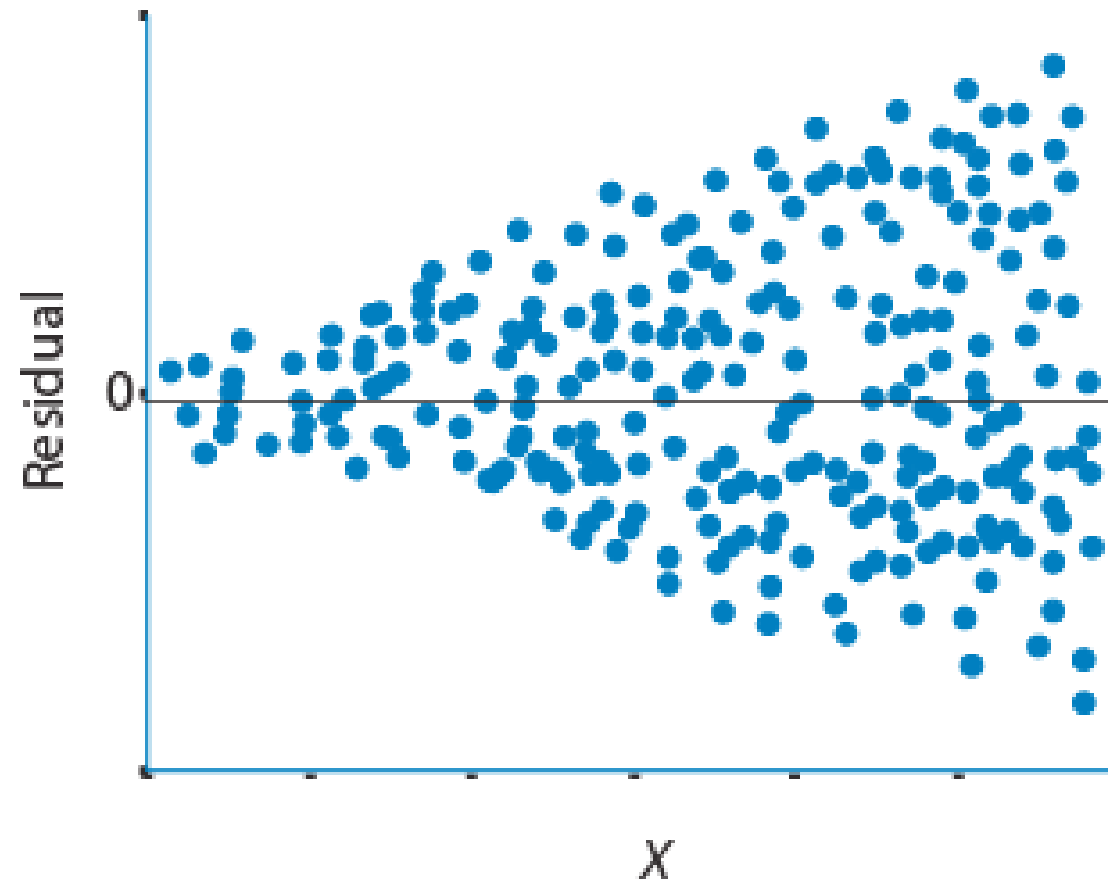
RESIDUAL PLOT

Example of linearity with equal variance



RESIDUAL PLOT

Example of linearity with unequal variance



RESIDUAL PLOT

Example of non-linearity with equal variance

